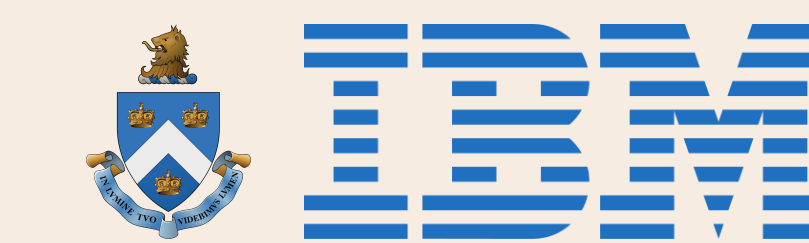


# TuxBot: Semantic-Aware Online OS Tuning with Large Language Models

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Paper



Code

## Problem

Bayesian & RL tuners treat hundreds of **coupled Linux knobs** as **isolated numbers**, blindly pushing live hosts into unsound regions.

**1,200+** tunable Linux knobs, often semantically coupled

## Motivation

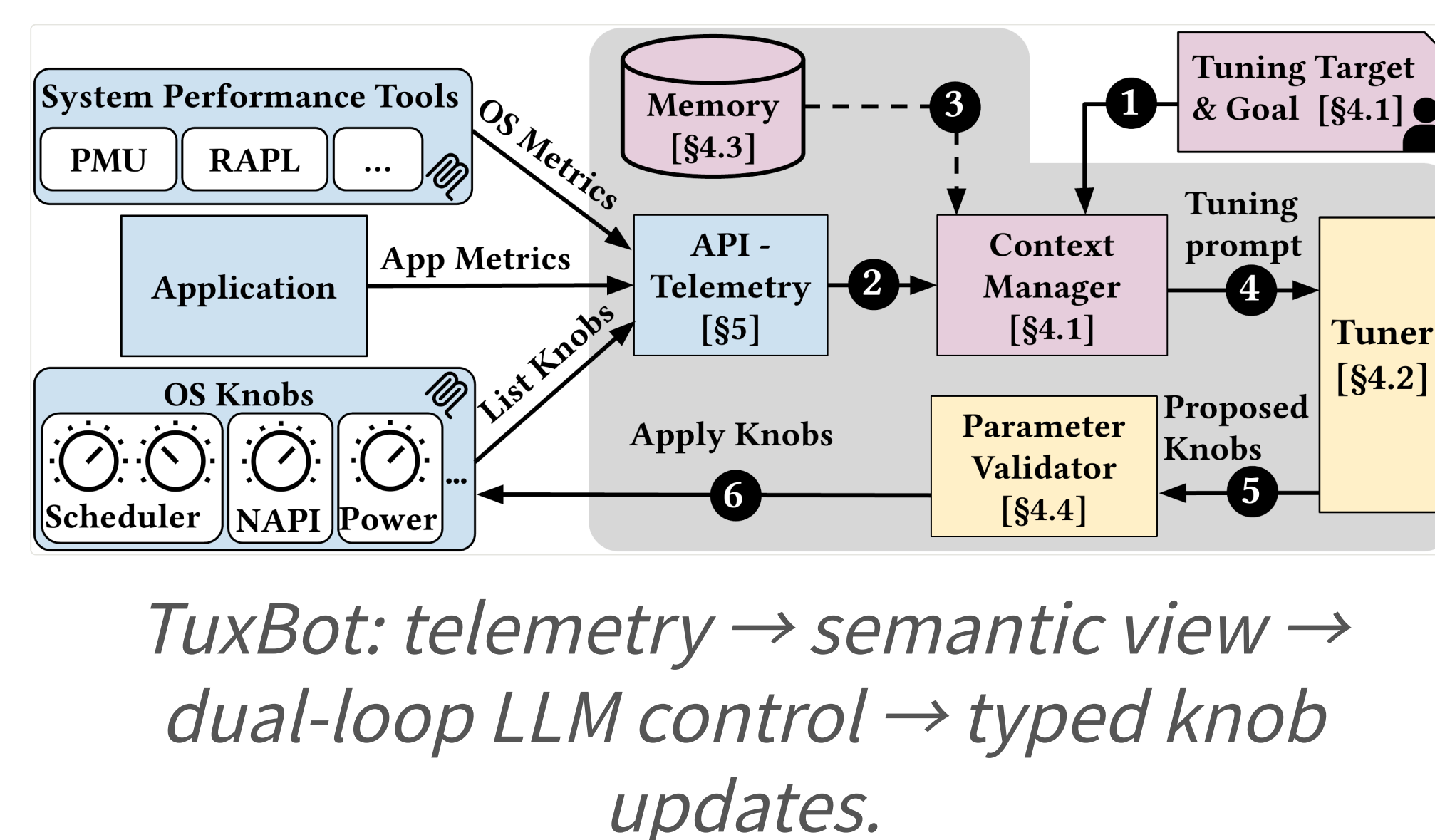
- Numerically valid configs turn **semantically unsound** once scheduler, power, memory & I/O interact on a live host.
- Hidden app metrics force **misleading proxies** (IPC, cache misses) that don't track real performance.

**>47×**  
tail-latency spike

**2×**  
worse p99 on proxies

**50%**  
p99 loss, 1→32 knobs

## Method



### 1 Semantic Reasoning

*knobs as policy*

Treats knobs as **one joint policy**, rejecting unsound configs up front:

**scheduler** **power** **memory** **I/O**

### 2 Cost-Aware Dual-Loop Control

*fast + deep*

1 **Observe** telemetry

2 **Instant** fast fix

3 **Reason** deeply

4 **Validate & Actuate**

Fast **Instant** + deep **Reasoning** loops — cost near **\$0.20/session**.

### 3 Explicit Cross-Session Memory

*warm start*

TuxBot records **summaries from prior runs** and retrieves them to warm-start new sessions, so tuning reuses workload-specific experience instead of restarting from scratch each time.

No re-learning — experience carries across runs.

### 4 Typed, Validated Actuation

*safe execution*

- Every LLM proposal passes a **typed, validated interface** before reaching the host.
- A knob metadata store covers **41** Linux knobs with domains and constraints.
- Keeps semantic reasoning **paired with safe execution** on live services.

**SAFE**

LLM proposals are type-checked before reaching the host.

## Key Results

| METHOD               | PERF.          | METRICS            |
|----------------------|----------------|--------------------|
| MLOS (best baseline) | baseline       | app-level          |
| TuxBot               | <b>+153.3%</b> | <b>system-only</b> |

## Headline Numbers

**+153%**  
avg. gain over MLOS (13 workloads)

**~\$0.20** LLM cost / session

**+93.7** pts, system-only

**41** Linux knobs co-tuned

## Takeaway

Treating OS knobs as a **coupled semantic system** — not isolated numbers — lets an LLM tuner deliver large, safe, cheap gains where classic tuners cannot.

*Semantic reasoning + typed actuation = safe, effective live tuning.*